
DIDB2210 • Database Applications

Department : Faculty of Information & Communication Technology
Course Name : Diploma in Multimedia and Software Engineering
Diploma in Information Technology
Diploma in Business Information Technology
Semester : 2
Commence Date : Week 8
Deadline Date : Week 14 & 15
Lecturers : Ms Mokhachane

Group Assignment 25%

PROJECT TITLE

DISTRIBUTED ONLINE WATER BILL MANAGEMENT DATABASE APPLICATION

PROJECT OVERVIEW AND OBJECTIVES

The project aims to develop a distributed internet database to manage customer information, billing data (including water consumption rates), and payment history for water bills. You are to design modules for sending bill notifications to customers in all districts of Lesotho. This project allows you to explore embedding queries to generate bills based on consumption and track payment statuses, potentially integrating with online payment gateways.

REQUIREMENTS

1. Design tables to manage customer information, billing rates, water usage and bills.
2. Write SQL queries to:
 - a) Calculate water bills based on consumption and applicable rates.
 - b) Track customer payment history and identify outstanding balances.
 - c) Generate reports on water usage patterns for different customer segments
3. Implement user accounts and secure login functionality.
4. Distribute this water billing application across **at least two (2) different databases**.
5. Consider incorporating payment processing using secure APIs (Note: This will require additional research on payment gateway APIs and security measures).

The project leverages various technologies such as web database applications, heterogeneous databases, and data integration to enhance convenience and improve the overall web database application experience for your local water service provider, Water and Sewerage Company (WASCO).

6. The application should serve as the main interface for accessing water usage and billing features and functionalities. This can be achieved through a combination of technologies, including Internet databases, secure APIs, embedded SQL and Online Analytical Processing. You can also include additional features to enhance the user experience such as reporting water leakages.

A. Design Issues for your consideration:

Your application should at least have these basic entities:

1. Customers (account number, name, address)
2. Billing Rates (rate tier, usage range, cost per unit)
3. Electricity Usage (account number, month, meter reading)
4. Bills (bill ID, account number, month, total amount due, payment status)

*Additional entities are welcome as you deem necessary.

- B. Other considerations:
1. The application should serve as the main interface for accessing online water usage and billing features and functionalities. It should provide a user-friendly interface where users can browse their water consumption rates, pay their bills and track their balances.
 2. The application should keep history of water bill payments that the user made in the last few weeks and months (at your discretion).
 3. The heterogeneous database management systems should store copies and fragments of WASCO's database relations.
 4. The online water billing management database application should provide an interface for the user to access:
 - a. GUI for registered and unregistered customers to view available water and sewerage services
 - b. GUI for registered users to navigate their portals.
 - c. GUI for an administrator to add, edit and view bill rates, bill payments and customers.
 - d. GUI for branch manager to view summative information and insights about the daily, weekly, monthly, quarterly and yearly water usage and bills.
- C. For this assignment, analyse WASCO water usage and billing and the possibility of designing an appropriate online database application which you will develop. The objective is to use the Database Applications concepts to complete this project. These concepts cover:
- Database modelling
 - Web database development
 - Advanced SQL (Views, Access control, Embedded SQL, TCL, DDL, DML and DDL)
 - Distributed databases

EXPECTED OUTCOMES

Students are expected to come up with the complete online water billing management database application and model and apply all concepts which they learned in this course in their project. Some of the necessary parts that should be included in the document are as follows:

- Problem definition
- Requirements
- Database models
- Justification of chosen web application technologies
- Justification of chosen DBMS technologies

SUBMISSION INFORMATION

- You must submit your final assignment as a complete document by Week 14 to your relevant Google Classroom and present your complete hosted application on the final presentation day (Week 15).
- You should prepare 20 minutes presentations on your project.
- Share the responsibilities among team members and all group members must contribute in the presentation.
- Failure to attend the presentation session (Week 15) will lead to getting a zero for the respective presentation mark.

SPECIFIC INFORMATION

- This assignment is group work, and it carries 25% of the course total mark.
- Your system can be developed using any preferred web application technology and database management systems.

RULES AND REGULATIONS

- Each team member should be involved in each aspect of the assignment.
- You are **NOT** allowed to copy diagrams, code or design structures from other sources. These parts must be the original work of the group

MARKING CRITERIA

You will receive marks based on the following table (25% of total mark for module):

Heterogeneous distributed database creation and connectivity	10
Hosted app online	10
GUI	10
Embedded SQL and Advanced SQL	10
Online water billing functions and features	15
Report/Document and data models	25
Confidence	5
Audibility	5
Qs&As	5
Teamwork	5
Time management	5
Dress Code	5
Total	110

Expected Report Content:

1. Title of the System
2. Table of Contents
3. List of Figures
4. List of Tables
5. List of Abbreviations
6. Abstract
7. Introduction
 - 1.1 Problem Statement
 - 1.2 Problem Solving
 - 1.3 Objectives
 - 1.4 Scope & Constraint
8. Technologies Justification
 - 2.1 Web application technologies
 - 2.2 Database Management System
9. System Design
 - 3.1 Data Models
- 14 Conclusion
 - 4.1 Advantages of the System
 - 4.2 Future Enhancement of the System
 - 4.3 Potential Benefit
 - 4.4 Conclusion
- 15 References
- 16 Appendices